

Web VAPT Section 10 - OS Command Injection Validation

Professional documentation for controlled OS command execution testing, impact analysis, and safe remediation.

Enterprise Security Assessment Lab

PROJECT AREA	Web Application Vulnerability Assessment and Penetration Testing
PRIMARY TARGET	DVWA running in local lab environment
TESTING ENVIRONMENT	Kali Linux attack machine with Docker-based vulnerable target
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DOCUMENT TYPE	Individual Web VAPT Attack-Section PDF

AUTHORIZED LAB TESTING ONLY

1. Document Purpose

This individual PDF documents the Web VAPT attack section titled 'OS Command Injection Validation'. It is designed for the Enterprise Security Assessment Lab GitHub portfolio and describes the objective, tooling, commands, sanitized output, evidence mapping, security interpretation, remediation, and redaction requirements for this specific section.

This document is written as a public portfolio version. It does not contain live client data, production system details, unredacted credentials, session cookies, raw hash dumps, or destructive attack instructions.

Field	Details
Project Area	Web Application Vulnerability Assessment and Penetration Testing
Primary Target	DVWA running in a local lab environment
Testing Environment	Kali Linux attack machine with Docker-based vulnerable target
Testing Style	Reconnaissance -> Enumeration -> Manual validation -> Exploitation proof -> Evidence capture
Methodology Reference	OWASP Web Security Testing methodology, OWASP Top 10, and PTES-style workflow
Prepared By	Jagriti Banerjee

2. Section Overview

Item	Details
Section Number	10
Section Title	Web VAPT Section 10 - OS Command Injection Validation
Risk / Severity Style	Critical in real-world context
OWASP Mapping	OWASP A03 - Injection
Primary Evidence Count	1

2.1 Objectives

- Test whether user input is passed unsafely to OS-level commands.
- Validate command execution using non-destructive identity commands.
- Document real-world risk without adding persistence or destructive steps.
- Provide secure alternatives to shell execution.

2.2 Tools Used

Tool	Purpose / Use in this section
Browser	Manual verification and proof screenshots.
Burp Suite	HTTP interception and manual request testing.
DVWA command execution module	Used as part of the controlled Web VAPT workflow.

3. Methodology and Execution Workflow

The execution workflow follows a controlled lab approach: confirm authorization and target scope, run only the tools required for the section, save outputs, validate observations manually where applicable, capture screenshots, redact sensitive values, and document findings without exaggeration.

3.1 Command Reference

```
# Non-destructive lab payloads
127.0.0.1; id
127.0.0.1; whoami

# Sanitized observed output pattern
uid=<REDACTED> gid=<REDACTED> groups=<REDACTED>
<REDACTED_WEB_SERVER_USER>
```

3.2 Expected / Sanitized Output

Command injection was validated in the DVWA lab using non-destructive identity commands. The evidence shows OS-level command output from user-controlled input.

3.3 Validation Criteria

- The target must be the local DVWA or authorized lab environment only.
- The output must match the intended section objective.
- Evidence screenshots must clearly show the validation result or tool output.
- Any sensitive values must be blurred, cropped, removed, or replaced with placeholders.
- The section must not claim findings that are not supported by evidence.

4. Evidence Mapping

Evidence File	Evidence Type / Reason
17-command-injection-id-whoami-output.png	OS command execution proof.

5. Professional Security Analysis

OS Command Injection is among the highest-risk web vulnerabilities because it crosses from application compromise into host-level compromise.

This lab uses safe identity commands to prove impact without destructive actions.

A professional report should stop at proof-of-impact unless deeper exploitation is explicitly authorized.

5.1 Why This Section Matters

This section matters because professional VAPT is not only about running a tool. It requires a clear objective, controlled execution, evidence capture, correct interpretation, and practical remediation. For recruiters and reviewers, this PDF shows that the work was structured, documented, and aligned with real security methodology.

5.2 Common Mistakes to Avoid

- Uploading raw outputs that expose cookies, hashes, paths, or credentials.
- Claiming production-level impact without production context.

- Reporting scanner output as a confirmed vulnerability without validation.
- Using unclear screenshots that do not show the relevant result.
- Mixing unrelated phases in one evidence file without explanation.

6. Risk Analysis

- Arbitrary command execution.
- File disclosure.
- Reverse shell access if abused.
- Server compromise.
- Lateral movement from the web host.
- Data theft or service disruption.

6.1 Real-World Impact Statement

The real-world impact depends on the affected asset, authentication context, data sensitivity, privilege level, exposed functionality, and compensating controls. Because this project uses DVWA in a lab, severity is described as a real-world context estimate rather than a formal client CVSS score.

7. Remediation and Hardening Guidance

- Avoid shell execution with user input.
- Use safe library APIs instead of command-line utilities.
- Apply strict allowlists for allowed values.
- Escape arguments safely if execution cannot be avoided.
- Run web services with least privilege.
- Monitor suspicious process execution.

7.1 Verification After Remediation

- 1 Re-run the original test case in the lab or development environment.
- 2 Confirm that the previous vulnerable behaviour no longer appears.
- 3 Check server logs for blocked or rejected attempts.
- 4 Capture new evidence showing the fix or defensive control.
- 5 Update the report status from validated/open to fixed/retested if applicable.

8. GitHub Redaction and Publishing Checklist

Cookies, hashes, credentials, session IDs, shell paths, database output, and private details must be redacted before public upload.

Item	Action Before Upload
PHPSESSID / Cookies	Blur or replace with <REDACTED_COOKIE>.

Item	Action Before Upload
Password hashes	Blur or replace with <REDACTED_HASH>.
Cracked passwords	Do not publish; replace with <REDACTED_PASSWORD>.
Database dumps	Crop, blur, or summarize instead of uploading raw data.
Webshell paths/files	Do not upload shell source files; redact paths if needed.
Local usernames/paths	Blur if private or unnecessary.
Tool raw outputs	Upload only sanitized outputs or summary documents.

8.1 Safe Git Commands Before Push

```
git status
git diff --cached
git diff --cached | grep -iE "password|secret|token|cookie|PHPSESSID|hash|credential"
```

9. Conclusion

This document completes the individual Web VAPT PDF for Section 10. It documents the section objective, execution workflow, evidence mapping, professional interpretation, risk, remediation, and GitHub safety checks. The content is aligned to the actual Enterprise Security Assessment Lab Web VAPT evidence set and avoids unsupported production claims.

Legal and ethical notice: This documentation is for authorized lab testing only. Do not apply these techniques to public or third-party systems without explicit written permission.